

Can cybersecurity keep pace with modern digital risk?

A new 2026 investigation built from my first A-level dissertation—and the question that led towards safer product decisions.

Nathan Brown-Bennett · Revised July 2026

Nathan Brown - Bennett: An investigation into Cyber Security and whether it is secure for modern day use?

Extended Project Qualification

EPQ 2020

Nathan Brown-Bennett

EPQ Title: An investigation into Cyber Security and whether it is secure for modern day use?

Abstract

This project aims to explore and define security, apply, and differentiate it between information security (IT Sec), Information Communication Security (ICT Sec) and Cyber Security. In order to assess how secure Cyber Security is for modern day operations of data security and its effectiveness in comparison to other security methods, I look at how Cyber Security works and used to work in a field of expertise whilst modernisation, automation and security changes throughout the years have changed its function in comparison to other methods, focusing on why and whether cybersecurity is important in the world of business as well as daily use.

One question became a research-to-product thread



Security is managed harm reduction—not a guarantee

A security decision identifies something of value, the threats and weaknesses that could affect it, the likely consequence, and the controls proportionate to that risk.

A practical test

What are we protecting?
From which harm?
Who owns the decision?
How will we know the control works?
What happens when it fails?

CONFIDENTIALITY

Prevent unauthorised disclosure

INTEGRITY

Prevent unauthorised change

AVAILABILITY

Keep authorised access and service

RESILIENCE

Contain disruption and restore trust

CONTEXT

Add safety, privacy, authenticity and accountability where needed

The security fields overlap—but protect different boundaries

FIELD	PRIMARY BOUNDARY	WHAT IT HANDLES
Computer security	One device or computing system	OS, software, memory, storage, local data
IT security	An organisation's technology estate	Devices, identities, networks, cloud, applications, backups
Information security	Information in any form	Digital, paper or spoken information; confidentiality, integrity, availability
Cybersecurity	Harm through connected digital systems	Technology, information, people, services and physical processes

The labels clarify ownership. They should not create silos.

Roles are organised around outputs—not universal job titles

FIELD	TYPICAL ROLES	ACCOUNTABLE OUTPUTS
Computer	Endpoint / platform engineer OS specialist · malware analyst	Hardened builds · patch baselines endpoint controls · findings
IT	Security analyst · IAM specialist network / cloud security engineer	Managed access · architecture monitoring · backup · recovery
Information	Security manager · risk specialist governance lead · asset owner	Classification · risk treatment policy · assurance · audit evidence
Cyber	SOC analyst · incident responder threat intel · pentest · AppSec	Threat-led defence · detection containment · testing · improvement

Small organisations may combine every duty in one IT role; larger organisations may split one duty across several teams.

One stolen credential creates four connected questions

Scenario: a staff laptop is compromised through a stolen cloud credential.

COMPUTER

IT

INFORMATION

CYBER



Was the endpoint patched, hardened and able to isolate activity?

How were the identity, device, network and cloud service administered?

Which records were exposed, how sensitive were they, and what duties follow?

What was the connected attack path, wider harm, containment and recovery?

Clear handoffs close the gaps: technical evidence → access revocation → impact assessment → coordinated containment and learning.

The original instinct was useful; the diagnosis needed work

WHAT SURVIVED

- Perfect security is not operational
- Public understanding matters
- Automation is necessary at scale
- Cybersecurity connects technology, organisations and society

2020 submission

WHAT CHANGED

- Breaches do not prove defence is useless
- “Human error” is a system-design signal
- AI augments accountable professionals
- The original survey was exploratory, not representative

2026 standalone dissertation

The real gap is adoption—not a binary score

Exposure is widespread. Governance and preparation remain inconsistent.

43%

UK businesses identifying a breach or attack

204

nationally significant incidents handled by the NCSC

30%

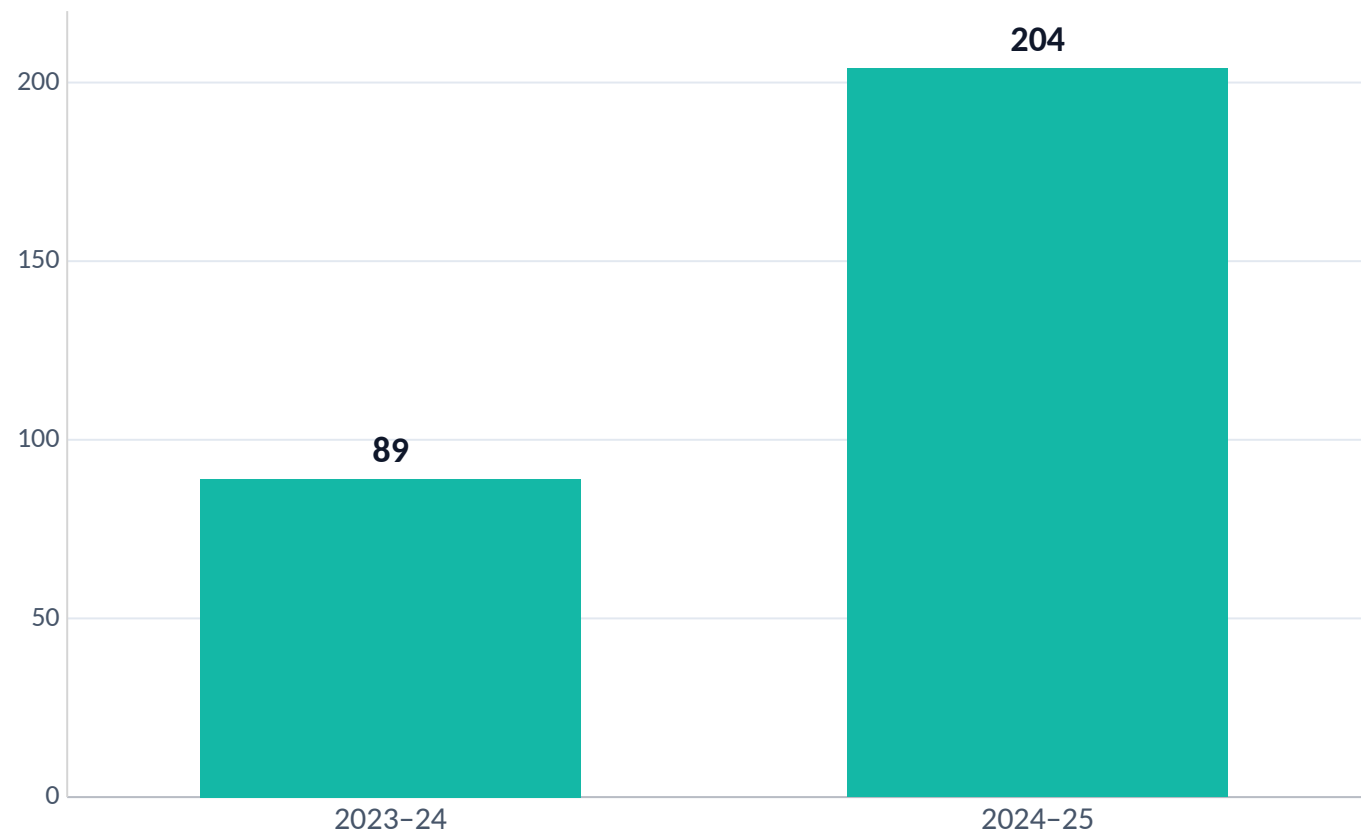
businesses completing a cyber-risk assessment

25%

businesses with a formal incident-response plan

Attack prevalence measures exposure and detection—not the absence of defensive value.

Nationally significant incidents rose sharply



2.3x

more nationally significant incidents handled than the previous year

Pressure increased—but the NCSC also states that many more attacks fail than succeed, and prepared organisations recover better.

A mistake is a design signal—not the end of the analysis

If one ordinary action can create catastrophic harm, examine the architecture around it.

REDUCE PRIVILEGE

One account should not expose everything.

USE SAFER DEFAULTS

Strong authentication and secure setup should be the easy path.

MAKE REPORTING SAFE

Early honesty should reduce harm—not trigger fear.

DESIGN RECOVERY

Contain failure and restore trusted operation.

Training matters. It is not a patch for an unsafe product.

AI changes both attack and defence—not accountability

DEFENCE

Triage alerts
Inspect code
Summarise threat intelligence
Spot unusual patterns

ATTACK

Scale convincing content
Automate reconnaissance
Search for weaknesses
Accelerate intrusion steps

NEW SYSTEM RISK

Training data
Model behaviour
Sensitive integrations
Over-trust in outputs

Accountable people use automation with evidence, audit trails, realistic testing and a way to challenge the result.

Effective cybersecurity connects six continuous functions

GOVERN

Own risk, priorities and suppliers

IDENTIFY

Know assets, dependencies and threats

PROTECT

Reduce likely harm with usable controls

DETECT

Know when something changes

RESPOND

Contain, decide and communicate

RECOVER

Restore trusted service and learn

The test is not whether a control exists. It is whether people can use it, failure is contained, and the organisation can demonstrate that it works.

The research became a safer decision workflow

The Device Provisioning Toolkit does not certify perfect safety.

It discovers devices, compares relevant security signals and produces a tailored hardening guide—turning evidence into a repeatable, auditable next action.

EPQ

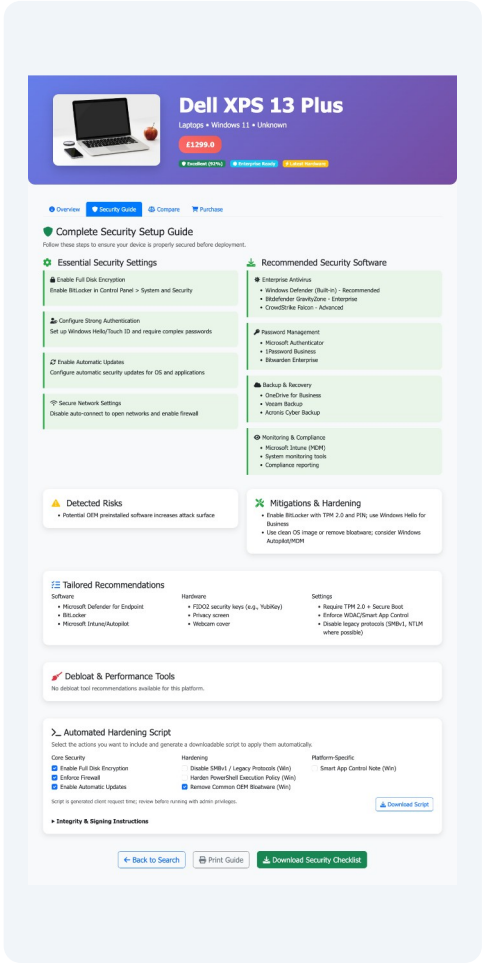
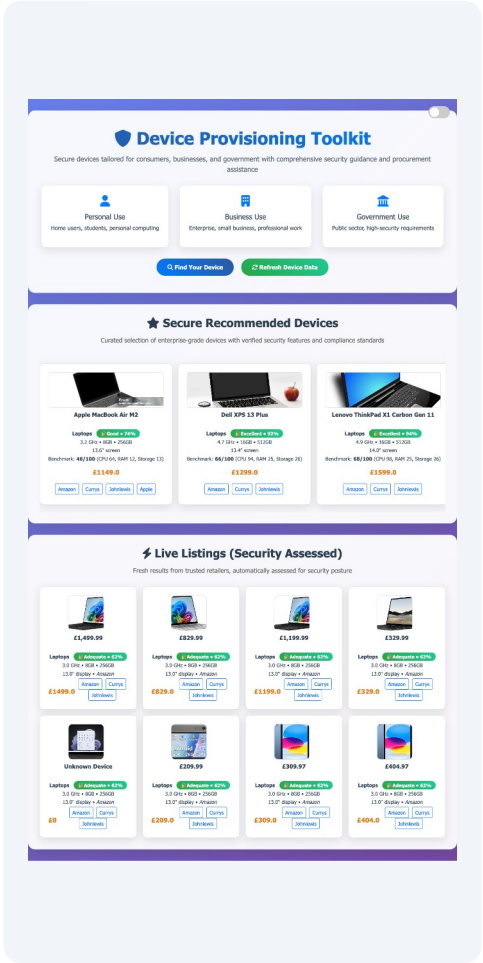
Security is never finished

UNDERGRADUATE

Device choices need clearer evidence

PRODUCT

Make the safer action easier



Cybersecurity can keep pace—when it is continuous, governed, usable and designed for recovery.

Next: “How do we decide what we provide?” and the Device Provisioning Toolkit